DRCPROP Tutorial

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This tutorial describes how the drcprop classes can be used to setup a simple optical systems for ray tracing.

Part 1

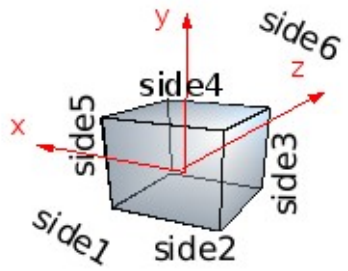
Let's start with the most simple example test_simple_bar2.cc

it is only a brick with the constructor

```
PndDrcOptBrick bar(17,17.5/2,100);
```

Dimensions 34x17.5x200 mm, the arguments are the half-lengths. Its center is positioned at 0,0,0. These are no physical units it could be mm,cm,m,km.

When you run doxygen to generate some class documentation, then you see there the picture of such a brick PndDrcOptBrick and the names of the surfaces.



Therefore the two surfaces side6 and side1 are the surfaces at z=+-100 and get now special functionality:

```
bar.Surface("side6")->SetReflectivity(PndDrcOptReflSilver());
bar.Surface("side1")->SetPixel();
```

side6 is now reflecting (reflection probability as function of wavelength defined in PndDrcOptReflSilver) and side1 is a pixel, if a photon hits this side the propagation is stopped.

Now a optical system is generated which in principle could contain many volumes which are connected (future step)

```
PndDrcOptDevSys opt_system;
opt_system.AddDevice(bar);
```

The manager holds all information about objects, one can add several optical systems. The idea was that if a detectors consists of many optical systems which are all the same, but different in position like the bars with lenses and mirrors in a barrel, they are all added to the manager

```
PndDrcOptDevManager* manager = new PndDrcOptDevManager();
manager->AddDeviceSystem(opt_system);
```

Now a filestream is created an connected to the manager

```
fstream geo;
geo.open("Geo.C",std::ios::out);
geo<<"{"<<endl;
geo<<"    TCanvas *c1 = new TCanvas(\"c1\"); "<<endl;
if ( ((TROOT*)gROOT)->GetVersionInt() < 51600)
      {
            geo<<"    TView *view = new TView(1);"<<endl;
      }
else
      {
            geo<<"    TView *view = TView::CreateView(1);"<<endl;
      }
geo<<"    view->SetRange(-50,-50,-50,50,50,50);"<<endl;
geo<<"    Int_t i;"<<endl;
geo<<"    view->SetView(0,90,90,i);"<<endl;
// the following command sets a flag within the manager and all photons from
// now on will be traced and can be plotted by calling within root
// .x Geo.C
// .x Screen.C
//
manager->Print(geo);
//
// the intention is to play around with routines.
```

if this

```
manager->Print(geo)
```

is not commented out each photons path element is written to a file together with the geometry of all objects. That is done in a stupid space consuming way (look to Geo.C which then is generated) that it is better for long photon lists to comment it out.

now create a particle with 0.69c 10mm under the bar and shoot in towards (0,1,1) through the bar

```
// create a list of photons in bar

XYZPoint  pos(0,-10,0);
XYZVector dir(0,1,1);
double   beta = 0.69;
bool photons_exist = manager->Cerenkov(pos,dir,beta); // generate photons
if (photons_exist) manager->Propagate();              // propagate photons
list<PndDrcPhoton> list_photon = manager->PhotonList();  // get list
```

if photons are generated by manager->Cerenkov(pos,dir,beta); then they are propagated by manager->Propagate(); and you can access the propagated photons by accessing a list

```
for(iph=list_photon.begin(); iph != list_photon.end(); ++iph)
```

```
        {
              if      ((*iph).Fate()==Drc::kPhotMeasured) icnt_measured++;
              else if ((*iph).Fate()==Drc::kPhotFlying)   icnt_flying++;
              // should never happen.
              else if ((*iph).Fate()==Drc::kPhotAbsorbed) icnt_absorbed++;
              else                                        icnt_lost++;
        }
```

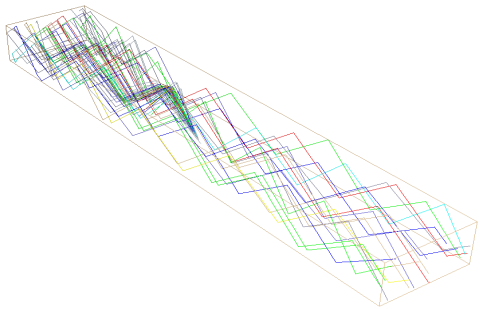
here it is simply the fate, but you could for example print out the coordinates, eg. (*iph).Position().X() and so on...

if you run the file you get the output

```
generated photons: 48
measured  photons: 18
absorbed  photons: 14
lost      photons: 16
```

Here, only 48 photons were generated since the speed of the particle just above the Cherenkov threshold. Absorbed in the quartz were 14 photons due to the photon attenuation lenghth of the quartz material or due to the reflectivity of the mirror. 16 photons were lost because they have not been internally reflected. The remainin 18 photons hit the surface “side1”.

A file Geo.C is generated which can be viewed with root:



One sees the generated Cherenkov photons reflectin in the +z direction beeing reflected and hitting “side1”. The color of the photons represent their wavelength: red is red, blue is blue, the color code one finds in

```
//-------------------------------------------------------------------------
int PndDrcPhoton::ColorNumber(double lambda) const
{
        // the colors were taken from wikipedia entry about colors.
        if      (lambda< 0.1) return 17; // grey
        if      (lambda> 740) return 50; // dark red
        else if (lambda> 625) return 2;  // red
        else if (lambda> 590) return 42; // orange
        else if (lambda> 565) return 5;  // yellow
        else if (lambda> 520) return 3;  // green
        else if (lambda> 500) return 7;  // cyan
        else if (lambda> 450) return 4;  // blue
        else if (lambda> 430) return 9;  // indigo
        else if (lambda> 380) return 39; // violet
        return 33; // blue grey
}
```

Part 2

In the next example we define a surface reflection probability for the quartz bars. Look to test_simple_bar2a.cc.
Including

```
#include "PndDrcOptReflGray.h"
```

the constructor

```
PndDrcOptReflGray refl;
refl.SetReflProb(0.99);
```

is available, the reflextivity is 99% and the outside surfaces of both bars get this quantity with eg.

```
bar1.Surface("side2")->SetReflectivity(refl);
```

It is important that all devices get names. With these names one can acess the objects. Internally within the classes all objects are cloned and have different pointers. Therefore, the names are needed to find these objects.

Instead of writing

```
PndDrcOptBrick bar2(17,17.5/2,100);
bar2.SetOptMaterial(PndDrcOptMatLithotecQ0());
bar2.SetName("bar2");
bar2.Surface("side1")->SetPixel();
bar2.Surface("side2")->SetReflectivity(refl);
bar2.Surface("side3")->SetReflectivity(refl);
bar2.Surface("side4")->SetReflectivity(refl);
bar2.Surface("side5")->SetReflectivity(refl);
```

the following is the same

```
PndDrcOptBrick bar2(bar1);
bar2.SetName("bar2");
bar2.Surface("side1")->SetPixel();
bar2.Surface("side6")->SetReflectivity(PndDrcOptReflNone());
// (one needs the definition #include PndDrcOptReflNone.h)
```

Now we shift the bars in the right position that “side1” of “bar1” faces “side6” of “bar2”.

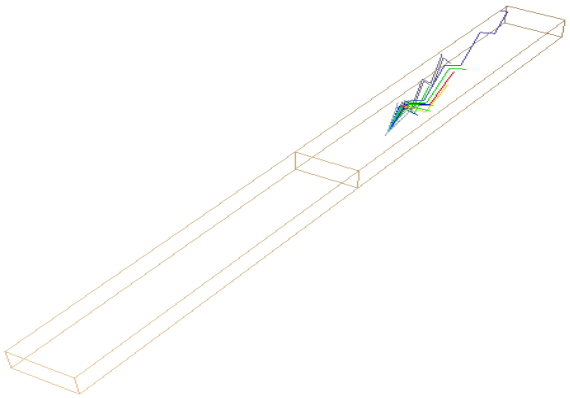
```
bar1.AddTransform(Transform3D(XYZVector(0,0, 100))); // shift by half length
bar2.AddTransform(Transform3D(XYZVector(0,0,-100)));
```

and after adding the to the optical system, we connect the corresponding surfaces (again using names):

```
// couple surface 1 of device 1 with surface 2 of device 2
//                        dev1   dev2    surf1     surf2
opt_system.CoupleDevice("bar1","bar2","side1","side6");
```

One can now play with the reflection probability and see how the number of measured photons change.

For a 50% reflection probability the photons do not reach the pixel side anymore.



Part 3

The optical active volumes are not restricted to boxes, they can consist of surfaces delimited by polygons. The surfaces within these polygons need not to be flat, they can even bend like eg. Lenses. The PndDrcOptBrick is internally a combination of 6 rectangular surfaces. One can do this also by hand. We will exercise this by constructing a prism shaped expansion box. This is the example in file test_simple_bar2b.cc.

```
// define an expansion volume with limitin points in z-y:
//
//              p3a
//          /       |
//        /         |
//      /           |              y
//    p2a           |              ^
//    |             |              |
//    p1a------p4a        z <--0
//
//  the left side will be connected to the quartz bar the right side will be the pixel.
//
//
//  seen from -z it looks like
//
//  p3a----p3b            Y
//  |          |          ^
//  |          |          |
//  |          |        x <--0
//  |          |
//  |          |
//  p4a----p4b
```

The coordinates of the points are given and 6 surfaces defined.

```
PndDrcSurfPolyFlat a1,a2,a3,a4,a5,a6;

a1.SetName("side+z");
a1.AddPoint(p4a);
a1.AddPoint(p4b);
a1.AddPoint(p3b);
a1.AddPoint(p3a);
a1.SetPrintColor(2);  // paint it in red
a1.SetPixel();        // pixel
```

The names are needed for later access of the object. It is important, that the sequence of added points goes either clock wise or counter clock wise around the surface. Otherwise:

```
a1.AddPoint(p4a);
a1.AddPoint(p3b); // swapped
a1.AddPoint(p4b); // swapped
a1.AddPoint(p3a);
```

would create a hour glass shaped surface (not wanted).

From the surfaces one constructs the volume. It is up to the programmer, that all sides are closed and the volume has no holes.

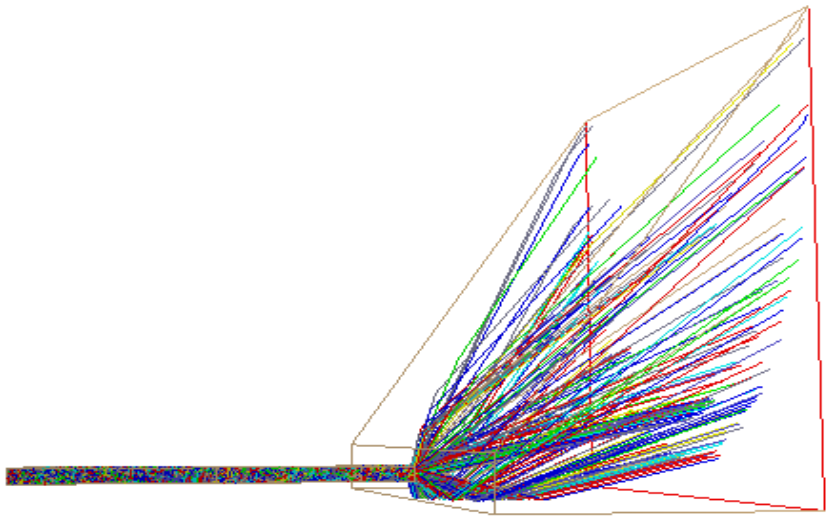
```
PndDrcOptVol ex_box;
PndDrcOptMatLithotecQ0 quartz;
ex_box.SetOptMaterial(quartz);
ex_box.AddSurface(a1);
ex_box.AddSurface(a2);
ex_box.AddSurface(a3);
ex_box.AddSurface(a4);
ex_box.AddSurface(a5);
ex_box.AddSurface(a6);
ex_box.SetName("expansion_box");

ex_box.Surface("side-z")->SetPixel();
```

The pixel is now the last surface in -z direction. The box is added to the optical system and the connection of the surfaces are

```
// couple surface 1 of device 1 with surface 2 of device 2
//                        dev1   dev2    surf1     surf2
opt_system.CoupleDevice("bar1","bar2",         "side1","side6");
opt_system.CoupleDevice("bar2","expansion_box","side1","side+z");
```

Running the code and “root Geo.C” shows



The example contains also a code section how to produce the coordinates of the photons which hit the backplane with colors representing the wavelength of the photons.

Run root Screen.C (Again the implemenatation is not very smart, but it works)